

Chapter 3 : Signed Measures and Differentiation

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Chapter 3 : Signed Measures and Differentiation

Problem 3.1. Prove Proposition 3.1.

Solution. Wlog suppose that ν omits the value $-\infty$, then if E_n is a sequence increasing sets we have that,

$$\nu\left(\bigcup_{n=1}^{\infty} E_n\right) = \nu(E_1) + \sum_{n \geq 2} \nu(E_n \setminus E_{n-1}) = \lim_{m \rightarrow \infty} \nu(E_1) + \sum_{n \leq m} \nu(E_n \setminus E_{n-1}) = \lim_{m \rightarrow \infty} \nu(E_m)$$

And if it was decreasing with $\nu(E_1)$ finite then, $E_1 \setminus E_n$ is increasing and we have

$$\nu(E_1) - \nu\left(\bigcap E_n\right) = \nu\left(E_1 \setminus \bigcap E_n\right) = \nu\left(\bigcup (E_1 \setminus E_n)\right) = \lim_{m \rightarrow \infty} \nu(E_1 \setminus E_m) = \nu(E_1) - \lim_{m \rightarrow \infty} \nu(E_m)$$

after which we can cancel out the finite term of $\nu(E_1)$. $\square$

Problem 3.2. If ν is a signed measure, E is ν -null iff $|\nu|(E) = 0$. Also, if ν and μ are signed measures, $\nu \perp \mu$ iff $|\nu| \perp \mu$ iff $\nu^+ \perp \mu$ and $\nu^- \perp \mu$.

Solution. If $|\nu|(E) = 0$ then $\nu^+(E) = \nu^-(E) = 0$ and as these are positive measures, by monotonicity see that $\nu^+(F) = \nu^-(F) = 0$ for any measurable $F \subseteq E$ making E ν -null. Now, if E was ν -null then $|\nu|(E) > 0$ would imply that $\nu^+(E)$ or $\nu^-(E)$ is strictly greater than 0 and if we assume the first to be as described then $\nu(E \cap N) = \nu^+(E \cap N) > 0$ which contradicts nullity of E w.r.t ν as $E \cap N \subseteq E$. The other case leads to a contradiction similarly, thus we must have $|\nu|(E) = 0$.

As a measurable set E is $|\nu|$ -null iff $|\nu|(E) = 0$ stemming from positivity we see that $\nu \perp \mu$ iff $|\nu| \perp \mu$ by the previous claim. We also see that $|\nu| \perp \mu$ iff $\nu^+, \nu^- \perp \mu$ as $|\nu|(E) = 0$ iff $\nu^-(E), \nu^+(E) = 0$, which proves the equivalence. $\square$

Problem 3.3. Let ν be a signed measure on $(X, \mathcal{M})$.

- $L^1(\nu) = L^1(|\nu|)$.
- If $f \in L^1(\nu)$, $|\int f d\nu| \leq \int |f| d|\nu|$.
- If $E \in \mathcal{M}$, $|\nu|(E) = \sup\{|\int_E f d\nu| : |f| \leq 1\}$.

Solution.

- As $L^1(\nu) := L^1(\nu^+) \cap L^1(\nu^-)$ we find that, $f \in L^1(\nu)$ implies $\int |f| d\nu^+, \int |f| d\nu^- < \infty$ and thus, $\int |f| d|\nu| = \int |f| d\nu^+ + \int |f| d\nu^- < \infty$ where we have used the following lemma: If μ_1, μ_2 are positive measures on the same sigma algebra then so is their sum and if $f \in L^+$ then $\int f d(\mu_1 + \mu_2) = \int f d\mu_1 + \int f d\mu_2$. For a proof, one can look at simple functions first where its trivial and then routinely extend to L^+ by MCT and approximations using simple functions. Now, if we had $f \in L^1(|\nu|)$ then as $\int f d|\nu| = \int |f| d\nu^+ + \int |f| d\nu^- < \infty$, both of these terms are individually finite which puts f in $L^1(\nu^+)$ and $L^1(\nu^-)$ proving the claim.
- This follows from the inequality $\int |f| d\nu = \int |f| d\nu^+ - \int |f| d\nu^- \leq \int |f| d\nu^+ + \int |f| d\nu^- = \int |f| d|\nu|$.

- c. Consider the hahn decomposition $P \cup N$ is usual notation leading upto the jordan decomposition, then letting $f = \chi_P - \chi_N$ we see that $\int f d\nu = \nu(P \cap E) - \nu(N \cap E) = \nu^+(E) + \nu^-(E) = |\nu|(E)$ so we have one direction of the inequality ($\leq$) and for the other direction, by part (b) we can say that $|f| \leq 1$ implies $|\int f d\nu| \leq \int 1 d|\nu| = |\nu|(E)$. so we are done.

□

Problem 3.4. If ν is a signed measure and λ, μ are positive measures such that $\nu = \lambda - \mu$, then $\lambda \geq \nu^+$ and $\mu \geq \nu^-$.

Solution. Consider the hahn decomposition $P \cup N$ in usual notation, leading upto the jordan decomposition. Then we have $\nu^+(E) = \nu(E \cap P) = \lambda(E \cap P) - \mu(E \cap P) \leq \lambda(E \cap P) \leq \lambda(E)$ and the other inequality follows similarly. □

Problem 3.5. If ν_1, ν_2 are signed measures that both omit the value $+\infty$ or $-\infty$, then $|\nu_1 + \nu_2| \leq |\nu_1| + |\nu_2|$. (Use Exercise 4.)

Solution. It can be easily shown that in this case $\nu_1 + \nu_2$ is also a signed measure. As all ν_1, ν_2 are finite, we can rearrange to get $\nu_1 + \nu_2 = \nu_1^+ - \nu_1^- + \nu_2^+ - \nu_2^- = (\nu_1^+ + \nu_2^+) - (\nu_1^- + \nu_2^-)$ which, by the previous problem implies $(\nu_1 + \nu_2)^+ \leq \nu_1^+ + \nu_2^+$ and $(\nu_1 + \nu_2)^- \leq \nu_1^- + \nu_2^-$ implying $|\nu_1 + \nu_2| \leq (\nu_1^+ + \nu_2^+) + (\nu_1^- + \nu_2^-) = (\nu_1^+ + \nu_1^-) + (\nu_2^+ + \nu_2^-) \leq |\nu_1| + |\nu_2|$. □

Problem 3.6. Suppose $\nu(E) = \int f d\mu$ where μ is a positive measure and f is an extended μ -integrable function. Describe the Hahn decompositions of ν and the positive, negative, and total variations of ν in terms of f and μ .

Solution. Let $P = \{f > 0\}$ and $N = \{f \leq 0\}$, then $P \cup N$ is clearly a hahn decomposition and we see that $\nu_+(E) = \int_{E \cap P} f d\mu = \int_E f^+ d\mu$ and similarly $\nu_-(E) = \int_E f^- d\mu$ and thus, $|\nu|(E) = \int_E |f| d\mu$. □

Problem 3.7. Suppose that ν is a signed measure on $(X, \mathcal{M})$ and $E \in \mathcal{M}$.

- $\nu^+(E) = \sup\{\nu(F) : F \in \mathcal{M}, F \subset E\}$ and $\nu^-(E) = -\inf\{\nu(F) : F \in \mathcal{M}, F \subset E\}$.
- $|\nu|(E) = \sup\{\sum_{j=1}^n |\nu(E_j)| : n \in \mathbb{N}, E_1, \dots, E_n \text{ are disjoint, and } \bigcup_{j=1}^n E_j = E\}$.

Solution.

- $F \subseteq E$ implies that $\nu(F) \leq \nu^+(F) \leq \nu^+(E)$ and $\nu(E \cap P) = \nu^+(E)$ with $E \cap P \subseteq E$ proving one part, the other follows similarly.
- Considering a hahn decomposition $P \cup N$ leading to a jordan decomposition, we see that $|\nu|(E) = \nu^+(E) + \nu^-(E) = |\nu(E \cap P)| + |\nu(E \cap N)|$ as $\nu^+(E) = \nu(E \cap P) \geq 0$ and $\nu^-(E) = -\nu(E \cap N) \geq 0$ proving one direction of the equality. For the other, it can be easily proved that $|\nu(E)| \leq |\nu|(E)$ and thus, $\bigcup_{j=1}^n E_j = E$ (disjoint union) implies $\sum_{j=1}^n |\nu(E_j)| \leq \sum_{j=1}^n |\nu|(E_j) = |\nu|(E)$ so we are done.

□

Problem 3.8. $\nu \ll \mu$ iff $|\nu| \ll \mu$ iff $\nu^+ \ll \mu$ and $\nu^- \ll \mu$.

Solution. If $\nu \ll \mu$ then $\mu(E) = 0 \implies \nu(E) = 0$ and as μ is taken positive we must have that $\mu(E \cap P), \mu(E \cap N) = 0$ where $P \cup N$ is the hahn decomposition for ν , this implies that $\nu(E \cap P), \nu(E \cap N) = 0$ so $\nu^\pm(E) = 0$ and we have that $|\nu|(E) = 0$ as well proving $|\nu| \ll \mu$. As $|\nu|(E) \geq |\nu(E)|$ we have that $|\nu| \ll \mu$ implies $\nu \ll \mu$ and hence $\nu \ll \mu$ iff $|\nu| \ll \mu$. If $\nu^+ \nu^- \ll \mu$ then $\mu(E) = 0$ clearly implies $|\nu|(E) = 0$ and if we have $|\nu| \ll \mu$ then $\mu(E) = 0$ implies $\nu^+(E) + \nu^-(E) = 0$ and thus $\nu^+(E) = \nu^-(E) = 0$ proving $\nu^+, \nu^- \ll \mu$ and thus all these conditions are equivalent. $\square$

Problem 3.9. Suppose $\{\nu_j\}$ is a sequence of positive measures. If $\nu_j \perp \mu$ for all j , then $\sum_1^\infty \nu_j \perp \mu$; and if $\nu_j \ll \mu$ for all j , then $\sum_1^\infty \nu_j \ll \mu$.

Solution. Say E_j is ν_j -null and E_j^c is μ -null for all j , then we see that, let $E = \bigcap E_j$, then we clearly see that E is $\sum_1^\infty \nu_j$ -null and we claim that, $E^c = \bigcup E_j^c$ is μ -null. For any measurable $F \subseteq E^c$, we can write $F = \bigcup_{j=1}^\infty (F \cap (E_j^c \setminus \bigcup_{i \neq j}^\infty E_i^c))$ is a disjoint union and

$$\mu(F) = \sum_{j=1}^\infty \mu(F \cap (E_j^c \setminus \bigcup_{i \neq j}^\infty E_i^c)) = 0$$

as $(F \cap (E_j^c \setminus \bigcup_{i \neq j}^\infty E_i^c)) \subseteq E_j^c$. $\square$

Problem 3.10. Theorem 3.5 may fail when ν is not finite. (Consider $d\nu(x) = dx/x$ and $d\mu(x) = dx$ on $(0, 1)$, or $\nu =$ counting measure and $\mu(E) = \sum_{n \in E} 2^{-n}$ on $\mathbb{N}$.)

Solution. We choose the second counterexample. Clearly we have, $\nu \ll \mu$ but for any $\delta > 0$ we see that there exists a E_δ with $\mu(E_\delta) < \delta$ but $\nu(E_\delta) \geq 1$ which just reads “ E is nonempty” and as the sum $\sum_{n \in \mathbb{N}} 2^{-n}$ converges absolutely we can find such a E_δ always. $\square$

Problem 3.11. Let μ be a positive measure. A collection of functions $\{f_\alpha\}_{\alpha \in A} \subset L^1(\mu)$ is called uniformly integrable if for every $\epsilon > 0$ there exists $\delta > 0$ such that $|\int_E f_\alpha d\mu| < \epsilon$ for all $\alpha \in A$ whenever $\mu(E) < \delta$.

- Any finite subset of $L^1(\mu)$ is uniformly integrable.
- If $\{f_n\}$ is a sequence in $L^1(\mu)$ that converges in the L^1 metric to $f \in L^1(\mu)$, then $\{f_n\}$ is uniformly integrable.

Solution.

- Consider $A = \{1, \dots, n\}$; for any $\epsilon > 0$, as $f_j d\mu \ll \mu$ where $f_j d\mu$ are finite measures, we can find $\delta_j > 0$ such that $|\int_E f_j d\mu| < \epsilon$ for all $j \in A$ whenever $\mu(E) < \delta_j$ so taking $\delta = \min_j \delta_j$ suffices.
- As f_n converges in L^1 , it is cauchy in this metric too so for any $\epsilon > 0$ we can find N such that $m, n \geq N \implies |\int_E f_m - \int_E f_n| \leq \int |f_m - f_n| < \epsilon$ and by the previous problem, as $\{f_1, \dots, f_N\}$ are uniformly integrable, we can find $\delta > 0$ such that, $|\int_E f_j| < \epsilon < 2\epsilon$ for $1 \leq j \leq N$ whenever $\mu(E) < \delta$ and for $n > N$,

$$\left| \int_E f_n d\mu \right| \leq \left| \int_E f_N d\mu \right| + \left| \int_E f_n - f_N d\mu \right| < \epsilon + \epsilon = 2\epsilon$$

so we are done.

□

Problem 3.12. For $j = 1, 2$, let μ_j, ν_j be σ -finite measures on $(X_j, \mathcal{M}_j)$ such that $\nu_j \ll \mu_j$. Then $\nu_1 \times \nu_2 \ll \mu_1 \times \mu_2$ and

$$\frac{d(\nu_1 \times \nu_2)}{d(\mu_1 \times \mu_2)}(x_1, x_2) = \frac{d\nu_1}{d\mu_1}(x_1) \frac{d\nu_2}{d\mu_2}(x_2).$$

Solution. Suppose $\mu_1 \times \mu_2(E) = 0$ then we see that $0 = \int \mu_1(E_x) d\mu_2(x)$ implying $\mu_1(E_x) = 0$ (this in turn implies $\nu_1(E_x) = 0$ here) for μ_2 -a.e. x so suppose its true for all x outside N with $\mu_2(N) = 0$ which implies $\nu_2(N) = 0$ so we have that $\nu_1 \times \nu_2(E) = \int \nu_1(E_x) d\nu_2(x) = 0$ as $\nu_1(E_x) = 0$ for ν_2 -a.e. x thus proving $\nu_1 \times \nu_2 \ll \mu_1 \times \mu_2$. Clearly these product measures are sigma finite too, so we can find a radon nikodyn derivative f i.e. $d(\nu_1 \times \nu_2) = f d(\mu_1 \times \mu_2)$ and also f_1, f_2 such that $f_j d\mu_j = \nu_j$ for $j = 1, 2$ and by fubini's we see that,

$$\begin{aligned} \int \chi_E(x, y) f_1(x) f_2(y) d(\mu_1 \times \mu_2(x, y)) &= \int \int f_1(x) \chi_{E_x}(y) f_2(y) d\mu_2(y) d\mu_1(x) \\ &= \int \nu_2(E_x) f_1(x) d\mu_1(x) = \int \nu_2(E_x) d\nu_1(x) = \nu_1 \times \nu_2(E) = \int \chi_E f(x, y) d(\mu_1 \times \mu_2(x, y)) \end{aligned}$$

which proves that $f(x, y) = f_1(x) f_2(y)$ for $\mu_1 \times \mu_2$ a.e. (x, y) so we have the equality treating them as equivalence classes. □

Problem 3.13. Let $X = [0, 1]$, $\mathcal{M} = \mathcal{B}_{[0,1]}$, m = Lebesgue measure, and μ = counting measure on $\mathcal{M}$.

- $m \ll \mu$ but $dm \neq f d\mu$ for any f .
- μ has no Lebesgue decomposition with respect to m .

Solution.

- As $\mu(E) = 0 \implies E = \emptyset$, $m \ll \nu$ holds. If we had $dm = f d\mu$ then we would have $m(\{x\}) = f(x) = 0$ for all x which is absurd as it implies $dm = 0$.
- If we had $\mu = \rho + \lambda$ where $\lambda \perp m$ and $\rho \ll m$ then if $[0, 1] = M \cup L$ where L, M are m, λ -null respectively then as $m(L) = 0$ we have that $m(M) = 1$ and thus, $\infty = \mu(M) = \rho(M) + \mu(M) = 0 + 1 = 1$ which is not possible.

□

Problem 3.14. If ν is an arbitrary signed measure and μ is a σ -finite measure on $(X, \mathcal{M})$ such that $\nu \ll \mu$, there exists an extended μ -integrable function $f : X \rightarrow [-\infty, \infty]$ such that $d\nu = f d\mu$. Hints:

- It suffices to assume μ is finite and ν is positive.
- With these assumptions there exists $E \in \mathcal{M}$ that is σ -finite for ν and $\mu(E) \geq \mu(F)$ for all sets F that are σ -finite for ν .
- The radon-nikodym theorem applies on E . If $F \cap E = \emptyset$, then either $\nu(F) = \mu(F) = 0$ or $\mu(F) > 0$ and $|\nu(F)| = \infty$.

Solution. If this is true for positive ν and finite μ , then for sigma finite μ we can write $X = \bigcup_1^\infty E_n$ with E_n being μ -finite and mutually disjoint we have $d\nu_n = f_n d\mu_n$ (we safely set $f_n = 0$ outside E_n) where $\mu_n(E) := \mu(E \cap E_n)$ and $\nu_n(E) = \nu(E \cap E_n)$ as μ_n is finite and clearly $\nu_n \ll \mu_n$ still holds. Then, with $f = \sum_1^\infty f_n$ we would have $d\nu = f d\mu$. If ν was signed on top of that, then using [Problem 3.8](#) which states $\nu \ll \mu \implies \nu^+, \nu^- \ll \mu$ we can find f^+, f^- with $d\nu^- = f^- d\mu, d\nu^+ = f^+ d\mu$ we have $d\nu = d\nu^+ - d\nu^- = (f^+ - f^-)d\mu$ and $f^+ - f^-$ is clearly extended integrable as ν , by definition can only assume atmost one of the values $+\infty, -\infty$. We now prove it for this case. Consider the set $\{\mu(H) : H \in \mathcal{M}, H \text{ is } \nu \sigma\text{-finite}\}$, let s be its supremum, then $s < \infty$ as μ is finite and we can find a $\nu \sigma$ -finite sequence of sets F_n such that $\mu(F_n) \rightarrow s$, then we can clearly see that $\mu(\bigcup F_n) = s$ as $\bigcup F_n$ is clearly still $\nu \sigma$ -finite and we can show both sides of the equality easily. Setting $E = \bigcup F_n$ gives us a set as described in hint (b). The radon nikodym theorem applies on E clearly. If $F \cap E = \emptyset$ then if we have $\nu(F) > 0$ we must have $\mu(F) > 0$ as $\nu \ll \mu$ which forces $\nu(F) = \infty$ as otherwise the existence of $E \cup F$ contradicts the maximality of E and if we have $\mu(F) > 0$ then again we similarly have $\nu(F) = \infty$ such as to not contradict the maximality of E so we have proved the claim in hint (c). There are now two cases, first one being when $\nu(E^c) = \mu(E^c) = 0$, in this case we can apply radon-nikodym to get a derivative f on E and extend it to the rest of the set by setting it to be zero outside this, then we see that,

$$\nu(K) = \nu(K \cap E) + \underbrace{\nu(K \cap E^c)}_{\leq \nu(E^c)=0} = \int_{K \cap E} f d\mu = \int_K f d\mu$$

as ν was assumed to be positive and $\mu(E^c) = 0$ so we have $d\nu = f d\mu$ in general so we are done. In the other case where $\mu(E^c) > 0$ and $\nu(E^c) = \infty$ we can set f to be ∞ on E^c , then we see that $\nu(E \cap K) = \int_{E \cap K} f d\mu$ as usual and either $\nu(E^c \cap K) = \mu(E^c \cap K) = 0$ or $\nu(E^c \cap K) = \infty$ and $\mu(E^c \cap K) > 0$ so we see that $\int_{E^c \cap K} f d\mu$ and $\nu(E^c \cap K)$ are either both infinite or both 0 by part (c) of the hint as $E^c \cap K$ is disjoint from E , hence we have the equality,

$$\nu(K) = \nu(E \cap K) + \nu(E^c \cap K) = \int_{E \cap K} f d\mu + \int_{E^c \cap K} f d\mu = \int_K f d\mu$$

so we are done. □

Problem 3.15. A measure μ on $(X, \mathcal{M})$ is called decomposable if there is a family $\mathfrak{F} \subset \mathcal{M}$ with the following properties: (i) $\mu(F) < \infty$ for all $F \in \mathfrak{F}$; (ii) the members of $\mathfrak{F}$ are disjoint and their union is X ; (iii) if $\mu(E) < \infty$ then $\mu(E) = \sum_{F \in \mathfrak{F}} \mu(E \cap F)$; (iv) if $E \subset X$ and $E \cap F \in \mathcal{M}$ for all $F \in \mathfrak{F}$ then $E \in \mathcal{M}$.

- a. Every σ -finite measure is decomposable.
- b. If μ is decomposable and ν is any signed measure on $(X, \mathcal{M})$ such that $\nu \ll \mu$, there exists a measurable $f : X \rightarrow [-\infty, \infty]$ such that $\nu(E) = \int_E f d\mu$ for any E that is σ -finite for μ , and $|f| < \infty$ on any $F \in \mathfrak{F}$ that is σ -finite for ν . (Use Exercise 14 if ν is not σ -finite.)

Solution.

- a. It can be easily shown that $\mathfrak{F} = \{E_n : n \in \mathbb{N}\}$ works where E_n partition X while being μ -finite.

- b. Let $\mathfrak{F}$ be as described in the problem. As $\mu(F) < \infty$ for any $F \in \mathfrak{F}$ we can find a measurable f_F on F such that $\nu(E \cap F) = \int_{E \cap F} f_F d\mu$ for all $E \in \mathcal{M}$ by the previous problem and if F is σ -finite for ν we can clearly force $|f_F| < \infty$ (in the proof presente in the book we can take all f_j 's which are zero outside A_j to be real valued everywhere as shown in case I which makes their sum real valued too as the A_j are disjoint.) as in the proof of the radon-nikodym theorem. We claim that $f := f_F$ on $F \in \mathfrak{F}$ is measurable and it has the properties needed. For any $E \in \mathcal{B}_{\mathbb{R}}$ we see that $f^{-1}(E) = \bigcup_{F \in \mathfrak{F}} f^{-1}(E) \cap F$ and each $f^{-1}(E) \cap F = f|_F^{-1}(E) \cap F = f_F^{-1}(E) \cap F \in \mathcal{M}$ as f_F is measurable which implies that $f^{-1}(E)$ by property (iv). Now, for any E that is σ -finite for μ , suppose $E = \bigcup E_n$ (disjoint union) with $\mu(E_n) < \infty$, then we see that $\mu(E_n) = \sum_{\mathfrak{F}} \mu(E_n \cap F) < \infty$ and it is well known that only atmost countably many $\mu(E_n \cap F)$ are nonzero, so we can find $\{F_m\} \subseteq \mathfrak{F}$ such that $\mu(E_n) = \sum_{m=1}^{\infty} \mu(E_n \cap F_m)$ and thus we also see that $\mu(E_n \setminus (\bigcup_m (E_n \cap F_m))) = 0$ which, by absolute continuity implies $\nu(E_n) = \nu(\bigcup_m (F_m \cap E_n)) = \sum_{m=1}^{\infty} \mu(E_n \cap F_m) = \sum_{m=1}^{\infty} \int_{E_n \cap F_m} f d\mu = \int_{E_n} f d\mu$ and thus, $\nu(E) = \sum_1^{\infty} \nu(E_n) = \sum_1^{\infty} \int_{E_n} f d\mu = \int_E f d\mu$.

□

Problem 3.16. Suppose that μ, ν are measures on $(X, \mathcal{M})$ with $\nu \ll \mu$, and let $\lambda = \mu + \nu$. If $f = d\nu/d\lambda$, then $0 \leq f < 1$ λ -a.e. and $d\nu/d\mu = f/(1 - f)$.

Solution. (note: we treat the derivatives as equivalence classes here.) Firstly, it is obvious that $f \geq 0$ λ -a.e. We have $\int_E d\nu = \nu(E) = \int_E f d(\mu + \nu) \leq \int_E d\nu + \int_E d\mu = \int_E d(\mu + \nu)$, it can then be easily shown that $f \leq 1$ λ -a.e. If we had $f = 1$ over E with $\lambda(E) = \nu(E) + \mu(E) > 0$ then we see that, $\nu(E) = \int_E f d\lambda = \lambda(E)$ which implies that $\mu(E) = 0$ but then as $\nu \ll \mu$ we have $\nu(E) = 0$ too contradicting non-nullity, this proves that $f < 1$ λ -a.e. too so we have proven the first claim. We clearly have $\nu \ll \mu \ll \lambda$ and $\lambda \ll \mu$. Thus, $g = d\mu/d\lambda$ exists and by Proposition 3.9 we see that $f = d\nu/d\lambda = (d\nu/d\mu)(d\mu/d\lambda) = g(d\nu/d\mu)$ and substituting these in we get $d\nu/d\mu = f/g = f/(1 - f)$ as $f + g = d\nu/d\lambda + d\mu/d\lambda = d(\mu + \nu)/d\lambda = d\lambda/d\lambda = 1$.

□

Problem 3.17. Let $(X, \mathcal{M}, \mu)$ be a σ -finite measure space, $\mathcal{N}$ a sub- σ -algebra of $\mathcal{M}$, and $\nu = \mu|_{\mathcal{N}}$ is σ -finite (This condition is missing in the book statement, without which this is false: Consider the lebesgue measure on $\mathbb{R}$ as the measure space and take $\mathcal{N} = \{\emptyset, \mathbb{R}\}$, $f = \chi_{[0,1]}$.) If $f \in L^1(\mu)$, there exists $g \in L^1(\nu)$ (thus g is $\mathcal{N}$ -measurable) such that $\int_E f d\mu = \int_E g d\nu$ for all $E \in \mathcal{N}$; if g' is another such function then $g = g'$ ν -a.e. (In probability theory, g is called the conditional expectation of f on $\mathcal{N}$.)

Solution. As $f \in L^1$ we have that $f d\mu$ is a signed measure and it is clearly still a measure restricted to $\mathcal{N}$ where we see that $(f d\mu)|_{\mathcal{N}} \ll \nu = \mu|_{\mathcal{N}}$ and as $\mathcal{N}$ is sigma finite for ν we can apply [Problem 3.14](#) to deduce existence of some g such that $(f d\mu)|_{\mathcal{N}} = g d\nu$ which implies $\int_E f d\mu = \int_E g d\nu$ for all $E \in \mathcal{N}$.

□

Problem 3.18. Prove Proposition 3.13c.

Solution. Let g be as in the definition ($d\nu = g d\mu$ with μ being positive) of $|\nu|$ then we see that $|f| d|\nu| = |f| |g| d\mu$ and we also have $|f| d\nu = |f| g d\mu$ and we have that,

$$\left| \int f d\nu \right| = \left| \int f g d\mu \right| \leq \int |f g| d\mu = \int |f| d|\nu|$$

□

Problem 3.19. If ν, λ are complex measures and μ is a positive measure, then $\nu \perp \mu$ iff $|\nu| \perp |\mu|$, and $\nu \ll \lambda$ iff $|\nu| \ll \lambda$.

Solution. If $\nu \perp \mu$ then by definition $\nu_a \perp \mu_b$ where $a, b = i, r$ following the notation in the book. By [Problem 3.2](#) this implies that $|\nu_a| \perp |\mu_b|$ where $a, b = i, r$ and using [Problem 3.9](#) twice we see that $|\nu_i| + |\nu_r| \perp |\mu_r| + |\mu_i|$. We see that $\nu \ll |\nu_i| + |\nu_r|$ as $\nu_a \ll |\nu_i| + |\nu_r|$ where $a = i, r$ and similarly for μ which implies that $\nu \perp |\mu_i| + |\mu_r|$ and by a similar argument, $\nu \perp \mu$. Now if $|\nu| \perp |\mu|$ then, as $\nu \ll |\nu|$ and $\mu \ll |\mu|$ we have that $\nu \perp \mu$. □

Problem 3.20. If ν is a complex measure on $(X, \mathcal{M})$ and $\nu(X) = |\nu|(X)$, then $\nu = |\nu|$.

Solution. We see that $\nu(X) = |\nu|(X) = |\nu(E) + \nu(X \setminus E)| \leq |\nu(E)| + |\nu(X \setminus E)| \leq |\nu|(E) + |\nu|(X \setminus E) = |\nu|(X) = \nu(X)$ so all the inequalities must be equalities, which implies that $|\nu(E)| = |\nu|(E)$ as $|\nu(E)| \leq |\nu|(E)$ for all $E \in \mathcal{M}$. As we have $|\nu(E)| + |\nu(X \setminus E)| = |\nu(E) + \nu(X \setminus E)|$ which is real, those two complex numbers share the same argument (unless 0 but that case is trivial) and as $\nu(E) + \nu(X \setminus E) = \nu(X)$ is real too, they must have no complex part making them real which implies that, $\nu(E) = |\nu(E)| = |\nu|(E)$ proving the claim. □

Problem 3.21. Let ν be a complex measure on $(X, \mathcal{M})$. If $E \in \mathcal{M}$ define

$$\mu_1(E) = \sup \left\{ \sum_1^n |\nu(E_j)| : n \in \mathbb{N}, E_1, \dots, E_n \text{ disjoint}, E = \bigcup_1^n E_j \right\},$$

$$\mu_2(E) = \sup \left\{ \sum_1^\infty |\nu(E_j)| : E_1, E_2, \dots \text{ disjoint}, E = \bigcup_1^\infty E_j \right\},$$

$$\mu_3(E) = \sup \left\{ \left| \int_E f \, d\nu \right| : |f| \leq 1 \right\}.$$

Then $\mu_1 = \mu_2 = \mu_3 = |\nu|$. (First show that $\mu_1 \leq \mu_2 \leq \mu_3$. To see that $\mu_3 = |\nu|$, let $f = \overline{d\nu/d|\nu|}$ and apply Proposition 3.13. To see that $\mu_3 \leq |\nu|$, approximate f by simple functions.)

Solution. $\mu_1 \leq \mu_2$ is obvious and $\mu_2 \leq \mu_3$ holds as for any given E and any partition $\{E_n\}$ of E we can define

$$f = \sum_n \frac{\overline{\nu(E_n)}}{|\nu(E_n)|} \cdot \chi_{E_n}$$

where the sum is over all n for which $\nu(E_n) \neq 0$, then we clearly see that $|f| \leq 1$ and $\int_E f \, d\nu = \sum_1^\infty \int_{E_n} f \, d\nu = \sum_1^\infty |\nu(E_n)|$ as $|z| = \bar{z}z/|z|$ for nonzero z . Clearly $\mu_3(E) \leq \int_E d|\nu| = |\nu|(E)$ by Proposition 3.13c and with f as in the hint (this can be made to have absolute value atmost 1 everywhere after changing it on a null set, not affecting any of the integrals below) we see that,

$$\int_E \frac{\overline{d\nu}}{d|\nu|} d\nu = \int_E \frac{\overline{d\nu}}{d|\nu|} \frac{d\nu}{d|\nu|} d|\nu| = \int_E |f|^2 d|\nu| = \int_E d|\nu| = |\nu|(E)$$

as $|f| = 1$ $|\nu|$ -a.e. so we must have $\mu_3 = |\nu|$. Now, we can find a sequence of simple functions ϕ_n converging f with $|\phi_n| \leq |f| \leq 1$ and convergence can be made uniform here as f is bounded on E , this clearly implies that $\lim |\int_E \phi_n d\nu| = |\int_E f d\nu|$ and thus, as $|\int_E \phi_n d\nu| \leq \mu_1(E)$ due to linearity and the triangle inequality we must have $\lim_n |\int_E f d\nu| \leq \mu_1(E)$ too so we are done. $\square$

Problem 3.22. If $f \in L^1(\mathbb{R}^n)$, $f \neq 0$, there exist $C, R > 0$ such that $Hf(x) \geq C|x|^{-n}$ for $|x| > R$. Hence $m(\{x : Hf(x) > \alpha\}) \geq C'/\alpha$ when α is small, so the estimate in the maximal theorem is essentially sharp.

Solution. We are given that f is not 0 a.e. and thus, $\int_{\mathbb{R}} |f| d\mu > 0$ and using DCT, we see that there exists some $r > 0$ such that $K = \int_{B(0,r)} |f| d\mu > 0$. Then, for all x , as $B(0, r) \subseteq B(x, |x| + r)$ we have

$$Hf(x) \geq \frac{1}{m(B(x, |x| + r))} \int_{B(x, |x| + r)} |f| d\mu \geq \frac{1}{c \cdot (|x| + r)^n} \int_{B(0,r)} |f| d\mu \geq \frac{K}{(|x| + r)^n}$$

where $c = m(B(0, 1))$. Thus, we can find some $C > 0$ and $R > 0$ like in the statement. Hence, for small $\alpha > 0$ we have $\{x : Hf(x) > \alpha\} \supseteq \{x : C\alpha^{-1} \geq |x|^n > R^n\}$. From this we can deduce that, $m(\{x : Hf(x) > \alpha\}) \geq c \cdot (C\alpha^{-1} - R^n)$ which is atleast $C'\alpha^{-1}$ for some $C' > 0$ and small α . $\square$

Problem 3.23. A useful variant of the Hardy-Littlewood maximal function is $H^*f(x) = \sup\{\frac{1}{m(B)} \int_B |f(y)| dy : B \text{ is a ball and } x \in B\}$. Show that $Hf \leq H^*f \leq 2^n Hf$.

Solution. If $x \in B(p, r) = B$ then we must have that $B(p, r) \subseteq B(x, |x - p| + r)$, thus

$$\frac{1}{m(B)} \int_B |f| dm \leq \frac{c \cdot (|x - p| + r)^n}{c \cdot r^n} Hf(x) \leq 2^n Hf(x)$$

where $c = m(B(0, 1))$, as $|x - p| < r$ so we must have $H^*f \leq 2^n Hf$. $\square$

Problem 3.24. If $f \in L^1_{loc}$ and f is continuous at x , then x is in the Lebesgue set of f .

Solution. For all $\epsilon > 0$ we can find $\delta > 0$ such that $f(B(x, \delta)) \subseteq B(f(x), \epsilon)$, thus for all $0 < \delta' < \delta$ we must have $m(B(x, \delta'))^{-1} \int_{B(x, \delta')} |f(x) - f(y)| dy \leq \epsilon$ so it must converge to 0 implying $x \in L_f$. $\square$

Problem 3.25. If E is a Borel set in $\mathbb{R}^n$, the density $D_E(x)$ of E at x is defined as $D_E(x) = \lim_{r \rightarrow 0} \frac{m(E \cap B(r, x))}{m(B(r, x))}$ whenever the limit exists.

- Show that $D_E(x) = 1$ for a.e. $x \in E$ and $D_E(x) = 0$ for a.e. $x \in E^c$.
- Find examples of E and x such that $D_E(x)$ is a given number $\alpha \in (0, 1)$, or such that $D_E(x)$ does not exist.

Solution.

- Consider the borel measure $\nu(F) := \mu(E \cap F)$, then we have a lebesgue decomposition $\nu = 0 + \chi_E dm$ and Theorem 3.22. we have that $D_E(x) = \chi_E(x)$ for μ -a.e. x . from which the conclusion follows.

- b. Consider $E = \bigcup_{k=1}^{\infty} (B(0, 2^{-2k+1}) \setminus B(0, 2^{-2k}))$ and $x = 0$. Let $c = m(B(0, 1))$. Then we see that for $m \in \mathbb{N}$ we have

$$m(E \cap B(0, 2^{-2m+1})) = c \sum_{k=m}^{\infty} (2^{n(-2k+1)} - 2^{-2kn}) = c(2^n - 1) \frac{2^{-2mn}}{1 - 2^{-2n}}.$$

Dividing by $m(B(0, 2^{-2m+1})) = c2^{n(-2m+1)}$, the density along this sequence of radii is:

$$\frac{m(E \cap B(0, 2^{-2m+1}))}{m(B(0, 2^{-2m+1}))} = \frac{c(2^n - 1)2^{-2mn}/(1 - 2^{-2n})}{c2^{-2mn}2^n} = \frac{2^n}{2^n + 1}.$$

Now consider the sequence of radii $r = 2^{-2m}$. The ball $B(0, 2^{-2m})$ intersects E starting from index $k = m + 1$, so:

$$m(E \cap B(0, 2^{-2m})) = c \sum_{k=m+1}^{\infty} (2^{n(-2k+1)} - 2^{-2kn}) = c(2^n - 1) \frac{2^{-2n(m+1)}}{1 - 2^{-2n}}.$$

Dividing by $m(B(0, 2^{-2m})) = c2^{-2mn}$ yields:

$$\frac{m(E \cap B(0, 2^{-2m}))}{m(B(0, 2^{-2m}))} = \frac{c(2^n - 1)2^{-2mn}2^{-2n}/(1 - 2^{-2n})}{c2^{-2mn}} = \frac{1}{2^n + 1}.$$

Since $\frac{2^n}{2^n + 1} \rightarrow 2$ and $\frac{1}{2^n + 1} \rightarrow 0$ for all $n \geq 1$, the limit depends on the sequence of radii chosen, hence $D_E(0)$ does not exist. To find an example where $D_E(x) = \alpha \in (0, 1)$, choose $x = 0$. If $n \geq 2$, define E as a sector of $\mathbb{R}^n$ originating at 0. Let $A \subset S^{n-1}$ be a measurable set on the unit sphere such that its surface measure is $\alpha\sigma(S^{n-1})$ which we can do by considering the function $f : [-1, 1] \rightarrow \mathbb{R}$ defined by $f(t) := \sigma(\{x \in S^{n-1} : x_1 \leq t\})$ which can be showed to be continuous via lower and upper continuity, after which we can use the intermediate value theorem. Define $E = \{y \in \mathbb{R}^n \setminus \{0\} : y/|y| \in A\}$. For any $r > 0$, $m(E \cap B(0, r)) = \alpha m(B(0, r))$, meaning $D_E(0) = \alpha$ identically for all $r > 0$. If $n = 1$, define $E = \bigcup_{k=1}^{\infty} \left(\left[\frac{1}{k} - \frac{\alpha}{k(k+1)}, \frac{1}{k} \right] \cup \left[-\frac{1}{k}, -\frac{1}{k} + \frac{\alpha}{k(k+1)} \right] \right)$. For $r = 1/k$, $m(E \cap (-r, r)) = 2 \sum_{j=k}^{\infty} \frac{\alpha}{j(j+1)} = \frac{2\alpha}{k} = \alpha(2r)$. Bounding the measure for $r \in (\frac{1}{k+1}, \frac{1}{k})$ shows the ratio to α , hence $D_E(0) = \alpha$.

□

Problem 3.26. If λ and μ are positive, mutually singular Borel measures on $\mathbb{R}^n$ and $\lambda + \mu$ is regular, then so are λ and μ .

Solution. Define $\nu = \lambda + \mu$ and let $\lambda(A) = \mu(A^c) = 0$. For compact K we see that $\lambda(K) + \mu(K) = \nu(K) < \infty$, which implies that $\lambda(K), \mu(K) < \infty$. First, we show outer regularity. If $\lambda(E) = \infty$, the condition $\lambda(E) = \inf\{\lambda(U) : U \supseteq E, U \text{ open}\}$ is trivially satisfied by any open $U \supseteq E$. Assume $\lambda(E) < \infty$. We wish to find an open $U \supseteq E$ with $\lambda(U \setminus E) < \epsilon$. We split E into $E \cap A^c$ and $E \cap A$. Notice that $\nu(E \cap A^c) = \lambda(E \cap A^c) + \mu(E \cap A^c) = \lambda(E \cap A^c) + 0 \leq \lambda(E) < \infty$. Since $\nu(E \cap A^c) < \infty$ and ν is regular, there exists an open $U_1 \supseteq (E \cap A^c)$ with $\nu(U_1 \setminus (E \cap A^c)) < \epsilon/2$, which implies $\lambda(U_1 \setminus (E \cap A^c)) \leq \nu(U_1 \setminus (E \cap A^c)) < \epsilon/2$. For the remaining piece $E \cap A$, we have $\lambda(E \cap A) \leq \lambda(A) = 0$. We can write $A = \bigcup_{k=1}^{\infty} A_k$ where $A_k = A \cap B(0, k)$. As each A_k is bounded, $\nu(A_k) < \infty$, so there exist open sets $V_k \supseteq A_k$ with $\nu(V_k \setminus A_k) < \epsilon/2^{k+1}$. This

implies $\lambda(V_k) = \lambda(V_k \setminus A_k) + \lambda(A_k) < \epsilon/2^{k+1} + 0$. Taking $U_2 = \bigcup_{k=1}^{\infty} V_k$ gives an open set containing A (and thus $E \cap A$) with $\lambda(U_2) \leq \sum \lambda(V_k) < \epsilon/2$. Let $U = U_1 \cup U_2$. Then U is open, $U \supseteq E$, and $U \setminus E \subseteq (U_1 \setminus (E \cap A^c)) \cup U_2$. Thus, $\lambda(U \setminus E) \leq \lambda(U_1 \setminus (E \cap A^c)) + \lambda(U_2) < \epsilon/2 + \epsilon/2 = \epsilon$ so we have the second condition. The first condition is trivial. $\square$

Problem 3.27. Verify the assertions in Examples 3.25.

Solution. This is trivial. $\square$

Problem 3.28. If $F \in NBV$, let $G(x) = |\mu_F|((-\infty, x])$. Prove that $|\mu_F| = \mu_{T_F}$ by showing that $G = T_F$ via the following steps.

- From the definition of T_F , $T_F \leq G$.
- $|\mu_F(E)| \leq \mu_{T_F}(E)$ when E is an interval, and hence when E is a Borel set.
- $|\mu_F| \leq \mu_{T_F}$, and hence $G \leq T_F$.

Solution.

- For any $x \in \mathbb{R}$ and $x_0 < \dots < x_n = x$ we see that,

$$\sum_{k=1}^n |F(x_k) - F(x_{k-1})| = \sum_{k=1}^n |\mu_F((x_{k-1}, x_k])| \leq \sum_{k=1}^n |\mu_F|((x_{k-1}, x_k]) = |\mu_F|((x_0, x])$$

Which is clearly less than $G(x)$. This shows that $T_F \leq G$ as T_F is the supremum over all sums of this form.

- For open (we only need these) intervals, the inequality follows from the fact that $T_F(x) - T_F(y) \geq |F(x) - F(y)|$ for $x \geq y$ as we can take limits and say that, for $a < b$

$$\mu_{T_F}((a, b)) = T_F(b-) - T_F(a) \geq |F(b-) - F(a)| = |\mu_F((a, b))|$$

(keep in mind that all the one sided limits exist in this case.) We know by construction that $\mu_F = \mu_1 - \mu_2 + i(\mu_3 - \mu_4)$ for finite lebesgue stjelt measures μ_i as in chapter 1. Let $\mu_5 = \mu_{T_F}$, another lebesgue stjelt measure. Given borel E , by regularity we can find open sets $U_n^{(i)} \supseteq E$ such that $\mu_i(U_n^{(i)}) \rightarrow \mu_i(E)$ for all i . Notice that by the  theorem and monotonicity of μ_i , the sequence of open sets $V_n := \bigcap_i U_n^{(i)}$ have the same convergence property by monotonicity of measures and the sandwich property. As μ_i are all finite we see that $\mu_F(V_n) \rightarrow \mu_F(E)$. If $\bigcup I_n$ is a countable disjoint union of open intervals we see that $|\mu_F(\bigcup I_n)| \leq \mu_{T_F}(\bigcup I_n)$ by the triangle inequality and sigma additivity of measures. From this, we can conclude that $|\mu_F(V_n)| \leq \mu_{T_F}(V_n)$ for all n and we have the desired inequality by taking the limit as $n \rightarrow \infty$

- This can be proven by the previous part and [Problem 3.21](#) directly (look at μ_1) in the hyperlinked problem. From this we thus have that $G \leq T_F$ by considering left open rays.

Now that we have $G = T_F$ we must have that $\mu_G = \mu_{T_F}$ and μ_G is none other than $|\mu_F|$. $\square$

Problem 3.29. If $F \in NBV$ is real-valued, then $\mu_F^+ = \mu_P$ and $\mu_F^- = \mu_N$ where P and N are the positive and negative variations of F .

Solution. It is easily seen that, the positive and negative variations are nonnegative for NBV functions as $F(-\infty) = T_F(-\infty) = 0$. Now, we clearly have $\mu_F = \mu_P - \mu_N$ (this is true for h -intervals and the sets where this holds forms a sigma algebra due to these being finite measures) which proves that $\mu_P \geq \mu_F^+$ and $\mu_N \geq \mu_F^-$ by [Problem 3.4](#). Now, from page 103 we can also see that

$$P(x) = \sup \left\{ \sum_{k=1}^n [F(x_k) - F(x_{k-1})]^+ : x_0 < \dots < x_n = x \right\} + 0$$

as $F \in NBV$, now notice that for $a < b$, we have

$$[F(b) - F(a)]^+ = [\mu_F((a, b))]^+ = [\mu_F^+((a, b)) - \mu_F^-((a, b))]^+ \leq [\mu_F^+((a, b))]^+ = \mu_F^+((a, b))$$

From this we can easily see that $\mu_P((-\infty, x]) = P(x) \leq \mu_F^+((-\infty, x])$. But earlier we proved that $\mu_P \geq \mu_F^+$ so we must have equality above and hence $\mu_P = \mu_F^+$ can be easily shown to hold. $\square$

Problem 3.30. Construct an increasing function on $\mathbb{R}$ whose set of discontinuities is $\mathbb{Q}$.

Solution. Consider an enumeration $\{q_n\}$ of the rationals. Define $f(x) := \sum_{q_n < x} 2^{-n}$ then it is obvious that for rationals r we have $f(q_n+) - f(q_n-) \geq 2^{-n}$ making it discontinuous on the rationals. Now, if x is an irrational, let $\delta > 0$ be such that $q_1, \dots, q_m \notin (x - \delta, x + \delta)$, we can do this because x is irrational. In this case, we see that for $y \in (x - \delta, x + \delta)$

$$|f(x) - f(y)| = \sum_{q_n \in [x - \delta, x + \delta)} \frac{1}{2^n} \leq \sum_{n > m} \frac{1}{2^n} = \frac{1}{2^{m+1}}$$

Thus we have continuity at the irrationals as this can be made arbitrarily small for larger m and small enough $\delta > 0$. $\square$

Problem 3.31. Let $F(x) = x^2 \sin(x^{-1})$ and $G(x) = x^2 \sin(x^{-2})$ for $x \neq 0$, and $F(0) = G(0) = 0$.

- F and G are differentiable everywhere (including $x = 0$).
- $F \in BV([-1, 1])$, but $G \notin BV([-1, 1])$.

Solution. The first part is an exercise in calculus and the next follows from F' being bounded ($|F'| \leq 3$ to be a bit more specific) and $T_G(1) - T_G(n^{-1/2}) \gg n$ (this is not to be interpreted as $\infty - \infty$ but rather the total variation in the smaller interval $(n^{-1/2}, 1)$) $\square$

Problem 3.32. If $F_1, F_2, \dots, F \in NBV$ and $F_j \rightarrow F$ pointwise, then $T_F \leq \liminf T_{F_j}$.

Solution. For any $x \in \mathbb{R}$ and $\epsilon > 0$ consider some $x_0 < \dots < x_n = x$ such that,

$$\sum_{k=1}^n |F(x_k) - F(x_{k-1})| \geq T_F(x) - \epsilon$$

Then, by pointwise convergence we see that for large enough j ,

$$-\epsilon + \sum_{k=1}^n |F(x_k) - F(x_{k-1})| \leq \sum_{k=1}^n |F_j(x_k) - F_j(x_{k-1})| \leq T_{F_j}(x)$$

This shows that $T_F(x) - 2\epsilon \leq T_{F_j}(x)$ for large enough j and thus, $T_F(x) - 2\epsilon \leq \liminf T_{F_j}(x)$. As $\epsilon > 0$ was arbitrary we are done. $\square$

Problem 3.33. If F is increasing on $\mathbb{R}$, then $F(b) - F(a) \geq \int_a^b F'(t)dt$.

Solution. As we only care about things being defined a.e. we define $F_n(x) := n(F(x + 1/n) - F(x)) \geq 0$ and we see that, $F_n \rightarrow F'$ a.e. thus, by fatou's lemma we have

$$\int_a^b F' \leq \liminf \int_a^b F_n = \liminf_n \int_a^b F(x + 1/n) - F(x) dm(x)$$

If we set $G(x) := \int_a^x F(t)dm(t)$ then we can pass to rieman integrability (monotone functions are rieman integrable) and see that the last expression above is

$$\liminf_n \frac{G(b + 1/n) - G(a + 1/n) - (G(b) - G(a))}{1/n} = G'(b) - G'(a) = F(b) - F(a)$$

As $G'(t) = F(t)$ by the fundamental theorem of calculus, when F is continuous at a, b . Now, as F is continuous a.e. the set of points where it is continuous must be dense and we can thus find sequences $a_n \searrow a < b \nearrow b_n$ where F is continuous and thus,

$$F(b) - F(a) \geq F(b-) - F(a+) = \lim_{n \rightarrow \infty} F(b_n) - F(a_n) \geq \lim \int_{[a_n, b_n]} F' = \int_{(a, b)} F' = \int_a^b F'$$

where the last two equalities are due to the fact that $A \mapsto \int_A F'$ is a measure here ((a_n, b_n) increases to (a, b) and $m(\{a, b\}) = 0$). $\square$

Problem 3.34. Suppose $F, G \in NBV$ and $-\infty < a < b < \infty$.

a. By adapting the proof of Theorem 3.36, show that

$$\int_{[a, b]} \frac{F(x) + F(x-)}{2} dG(x) + \int_{[a, b]} \frac{G(x) + G(x-)}{2} dF(x) = F(b)G(b) - F(a-)G(a-)$$

b. If there are no points in $[a, b]$ where F and G are both discontinuous, then

$$\int_{[a, b]} F dG + \int_{[a, b]} G dF = F(b)G(b) - F(a-)G(a-).$$

Solution. gonna do this later $\square$

Problem 3.35. If F and G are absolutely continuous on $[a, b]$, then so is FG , and

$$\int_a^b (FG' + GF')(x)dx = F(b)G(b) - F(a)G(a).$$

Solution. It suffices to show that F being absolutely continuous implies F^2 is absolutely continuous. Indeed, if F is absolutely continuous then it is continuous too and thus is bounded on the compact set $[a, b]$, say $|F| \leq M$. If, $\epsilon > 0$ is arbitrary, we can find $\delta > 0$ with,

$$\sum_{i=1}^n (b_i - a_i) < \delta \implies \sum_{i=1}^n |F(b_i) - F(a_i)| < \frac{\epsilon}{M+1} \text{ for } a \leq a_1 < b_1 < \dots < b_n \leq b$$

Then,

$$\sum_{i=1}^n |F(b_i)^2 - F(a_i)^2| \leq M \sum_{i=1}^n |F(b_i) - F(a_i)| \leq \frac{\epsilon M}{M+1} < \epsilon$$

so F^2 is also absolutely continuous. Now, it is clear that F, G being absolutely continuous makes their product absolutely continuous too. From Theorem 3.35c we see that $(FG)' \in L^1([a, b], m)$ and the same also holds for F', G' . As they all exist a.e., they exist together a.e. as well and we can say that $(FG)' = FG' + F'G$ a.e. and again by the cited theorem,

$$F(b)G(b) - F(a)G(a) = \int_a^b (FG)'(x)dx = \int_a^b (FG' + F'G)(x)dx$$

□

Problem 3.36. Let G be a continuous increasing (we need G to be strictly increasing, otherwise consider constant functions.) function on $[a, b]$ and let $G(a) = c, G(b) = d$.

- If $E \subset [c, d]$ is a Borel set, then $m(E) = \mu_G(G^{-1}(E))$. (First consider the case where E is an interval.)
- If f is a Borel measurable and integrable function on $[c, d]$, then $\int_c^d f(y)dy = \int_a^b f(G(x))dG(x)$. In particular, $\int_c^d f(y)dy = \int_a^b f(G(x))G'(x)dx$ if G is absolutely continuous.
- The validity of (b) may fail if G is merely right continuous rather than continuous.

Solution.

- The case where this is an interval follows as shown, for $c \leq x < y \leq d$ we just have $\mu_G(G^{-1}([x, y])) = \mu_G([G^{-1}(x), G^{-1}(y)]) = G \circ G^{-1}(y) - G \circ G^{-1}(x) = y - x = m([x, y])$. We can easily show that the collection of sets E for which this works is a sigma algebra as the inverse commutes with the complement and unions, we thus have that the borel sigma algebra over $[c, d]$ is contained in this one which suffices.
- If $f = \chi_E$ ($E \in \mathcal{B}_{[c, d]}$) then, $\int_a^b \chi_E(G(x))dG(x) = \int_a^b \chi_{G^{-1}(E)}dG(x) = \mu_G(G^{-1}(E)) = m(E) = \int_c^d \chi_E dm$ and we can conclude by linearity of the integral and MCT as usual. If G is NBV, we have that $dG = G'dm$ (can be proven from 3.35c) and by Problem 2.14 we are done (split the integrable function into its two parts as usual.)
- In that case the range of f needn't be an interval in the first place. For example, take $f(x) := x\chi_{[0, 1/2)}(x) + (x+1)\chi_{[1/2, 1]}$ which is right continuous, increasing and maps $[0, 1]$ to $[0, 1/2) \cup [3/2, 2]$.

□

Problem 3.37. Suppose $F : \mathbb{R} \rightarrow \mathbb{C}$. There is a constant M such that $|F(x) - F(y)| \leq M|x - y|$ for all $x, y \in \mathbb{R}$ (that is, F is Lipschitz continuous) iff F is absolutely continuous and $|F'| \leq M$ a.e.

Solution. For the only if direction, absolute continuity follows directly from the definition by considering $\epsilon/M + 1$ instead of ϵ and for the derivative to exist a.e. (it is trivially bounded as needed when it does exist) we consider intervals $(n, n + 1]$ for $n \in \mathbb{Z}$. Clearly F is still absolutely continuous on this small interval and by Theorem 3.35c we again see that F' exists a.e. in $(n, n + 1]$ and thus it exists a.e. in $\mathbb{R}$ too as the union of countably many disjoint null sets is still null. To prove the if part, for $x < y$ consider F on the interval $[x, y]$. By Theorem 3.35c we can write $F(x) - F(y) = \int_x^y F'(t)dt$ again and by hypothesis we have,

$$|F(x) - F(y)| = \left| \int_x^y F'(t)dt \right| \leq \int_x^y |F'(t)|dt \leq M(y - x) = M|x - y|$$

The case of $x > y$ is no different. □

Problem 3.38. If $f : [a, b] \rightarrow \mathbb{R}$, consider the graph of f as a subset of $\mathbb{C}$, namely, $\{t + if(t) : t \in [a, b]\}$. The length L of this graph is by definition the supremum of the lengths of all inscribed polygons.

- a. Let $F(t) = t + if(t)$; then L is the total variation of F on $[a, b]$.
- b. If f is absolutely continuous, $L = \int_a^b [1 + f'(t)^2]^{1/2} dt$.

Solution.

- a. This is trivial, just remember how $|x + iy| = \sqrt{x^2 + y^2}$.
- b. If f is absolutely continuous then so is F and by problem 3.35c it can be shown that $\mu_F = F'dm$ on this interval and thus, $\mu_{T_F} = |F'|dm$ by [Problem 3.28](#), this shows that

$$L = T_F(b) - T_F(a) = \mu_{T_F}((a, b]) = \int_{(a, b]} |F'(t)|dm(t) = \int_a^b [1 + f'(t)^2]^{1/2} dm(t)$$

as $F'(t) = 1 + i \cdot f'(t)$ (when it exists.) □

Problem 3.39. If $\{F_j\}$ is a sequence of nonnegative increasing functions on $[a, b]$ such that $F(x) = \sum_1^\infty F_j(x) < \infty$ for all $x \in [a, b]$, then $F'(x) = \sum_1^\infty F'_j(x)$ for a.e. $x \in [a, b]$. (It suffices to assume $F_j \in NBV$. consider the measures μ_{F_j} .)

Solution. Keep in mind the following lemma : If $P_1, P_2, \dots$ hold a.e. then $\bigwedge P_i$ does too, the proof is left as an exercise. We can consider the functions $G_j(x) = F_j(x+) - F_j(a)$ and $G(x) = F(x+) - F(a)$ instead. These have the same properties as F_j and F , and we extend them to make them NBV as done before (they were trivially in $BV([a, b])$ already). Note that $G = \sum G_j$ everywhere, which follows from an application of the monotone convergence theorem with respect to the counting measure. Consider the

measures μ_G and μ_{G_j} , for which it holds that $\mu_G = \sum_j \mu_{G_j}$ since they agree on right-closed rays. Now, consider the Lebesgue-Radon-Nikodym decompositions $dG = g dm + d\lambda$ and $dG_j = g_j dm + d\lambda_j$. We have $\lambda_j \perp m$, and thus $\sum \lambda_j \perp m$ too by [Problem 3.9](#) (notice that the proof is still valid if ν_j are complex measures as well.) Since $\mu_G = \sum \mu_{G_j}$, we must have $dG = (\sum_j g_j) dm + \sum \lambda_j$. By the uniqueness criterion in the Lebesgue-Radon-Nikodym theorem, this implies $g = \sum g_j$ a.e. and $\lambda = \sum \lambda_j$. It is clear from the Lebesgue differentiation theorem that $g_j = G'_j$ a.e. and $g = G'$ a.e., so the lemma at the beginning implies that $G' = \sum G'_j$ a.e. If we now show that $F' = G'$ and $F'_j = G'_j$ a.e., then we would be done using the same lemma. Suppose p and q are such that $p = q$ a.e. and both are differentiable a.e. We want to show that $p' = q'$ a.e. We can easily find a set $A \subseteq [a, b]$ where $p = q$ and both are differentiable, such that $m([a, b] \setminus A) = 0$. Notice that A must be dense in this interval, so for any $x \in A$, we can find a sequence $x_n \in A \setminus \{x\}$ that tends to x . By the sequential criterion for limits, we have

$$p'(x) = \lim_{n \rightarrow \infty} \frac{p(x_n) - p(x)}{x_n - x} = \lim_{n \rightarrow \infty} \frac{q(x_n) - q(x)}{x_n - x} = q'(x)$$

for all $x \in A$, so we are done. $\square$

Problem 3.40. Let F denote the Cantor function on $[0, 1]$ (see §1.5), and set $F(x) = 0$ for $x < 0$ and $F(x) = 1$ for $x > 1$. Let $\{[a_n, b_n]\}$ be an enumeration of the closed subintervals of $[0, 1]$ with rational endpoints, and let $F_n(x) = F((x - a_n)/(b_n - a_n))$. Then $G = \sum_1^\infty 2^{-n} F_n$ is continuous and strictly increasing on $[0, 1]$, and $G' = 0$ a.e.

Solution. $\square$

Problem 3.41. Let $A \subset [0, 1]$ be a Borel set such that $0 < m(A \cap I) < m(I)$ for every subinterval I of $[0, 1]$.

- Let $F(x) = m([0, x] \cap A)$. Then F is absolutely continuous and strictly increasing on $[0, 1]$, but $F' = 0$ on a set of positive measure.
- Let $G(x) = m([0, x] \cap A) - m([0, x] \setminus A)$. Then G is absolutely continuous on $[0, 1]$, but G is not monotone on any subinterval of $[0, 1]$.

Solution. $\square$

Problem 3.42. A function $F : (a, b) \rightarrow \mathbb{R}$ ($-\infty \leq a < b \leq \infty$) is called convex if $F(\lambda s + (1 - \lambda)t) \leq \lambda F(s) + (1 - \lambda)F(t)$ for all $s, t \in (a, b)$ and $\lambda \in (0, 1)$.

- F is convex iff for all $s, t, s', t' \in (a, b)$ such that $s \leq s' < t'$ and $s < t \leq t'$,

$$\frac{F(t) - F(s)}{t - s} \leq \frac{F(t') - F(s')}{t' - s'}$$

- F is convex iff F is absolutely continuous on every compact subinterval of (a, b) and F' is increasing (on the set where it is defined).
- If F is convex and $t_0 \in (a, b)$, there exists $\beta \in \mathbb{R}$ such that $F(t) - F(t_0) \geq \beta(t - t_0)$ for all $t \in (a, b)$.

- d. (Jensen's Inequality) If $(X, \mathcal{M}, \mu)$ is a measure space with $\mu(X) = 1$, $g : X \rightarrow (a, b)$ is in $L^1(\mu)$, and F is convex on (a, b) , then

$$F\left(\int g \, d\mu\right) \leq \int F \circ g \, d\mu.$$

Solution.

□